


Figure 2.3 Michael Faraday and James Clerk Maxwell.

What was known about electricity and magnetism at the beginning of the eighteenth century consisted of little other than a few amusing sideshow tricks: glass rods which attract pieces of paper; magnets which repel and attract. The study of electricity and magnetism continued slowly throughout the eighteenth century and into the nineteenth, where we find Faraday working in a London laboratory full of bobbins, needles, knives and iron cages, exploring how electrical and magnetic things attract and repel. A good Newtonian, he tries to understand the force which acts between charged and magnetic things. But slowly, guided by his hands, in close contact with these objects, he is led to an intuition that will become the basis of modern physics. He ‘sees’ something new.

His intuition is this: we must not think of forces acting directly between distant objects, as Newton presumed. We must instead think that there exists an entity diffused throughout space, which is modified by electric and magnetic bodies and which, in turn, acts upon (pushes and pulls) the bodies. This entity, whose existence Faraday intuits, is today called the field.

What is it, then, a field? Faraday sees it as formed by bundles of very thin lines (infinitely thin), which fill space; an invisible, gigantic cobweb filling everything around us. He calls these lines ‘lines of force’ because, in some way, these lines ‘carry the force’: they transmit the electric and the magnetic